

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5- Coding Challenge	Assessment:	Assessment To 2 Coding Challenge
Weightage:	40% of CIA	Total Marks:	100
CO 5 Statement:	Analyze and apply clustering evaluation techniques such as the Elbow Method and Silhouette Score to determine the optimal number of clusters for effective data grouping and model selection. (BL4)		
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Section / Batch:	CSE AI 2024	Date:	26-08-2026

Code of Conduct

I_Y SRAVAN KUMAR (192472289)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____SRAVAN_____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

CO5-AT2 Coding Challenge

Optimal Number of Clusters using Elbow Method and Silhouette Score

Aim

To determine the optimal number of clusters using the **Elbow Method** and **Silhouette Score**, calculate the silhouette score from scratch, and select the most suitable value of k when the methods disagree.

Procedure

1. Create or load an unlabelled dataset.
2. Apply K-Means for different values of k.
3. Calculate **WCSS (Within-Cluster Sum of Squares)** for the Elbow Method.
4. Calculate the **Silhouette Score manually** without using `sklearn.metrics.silhouette_score`.
5. Compare the results of both methods.
6. If both methods suggest the same k, select that value.
7. If they disagree, compare the silhouette scores and choose the value giving better cluster separation while considering the elbow point.
8. Display the optimal number of clusters.

Pseudocode

START

Load the dataset

FOR each value of K from 2 to 10:

 Apply K-Means clustering

Calculate WCSS (Within-Cluster Sum of Squares)

FOR each data point:

Calculate average distance from
other points in the same cluster → a

Calculate average distance from
points in the nearest other cluster → b

Calculate Silhouette Score:

$$S = (b - a) / \max(a, b)$$

Calculate average Silhouette Score

Store WCSS and Silhouette Score

END FOR

Plot K versus WCSS

Identify the Elbow point

Select K using the Elbow Method

Select K with the highest Silhouette Score

IF both methods select the same K:

Select that K

ELSE:

Compare Silhouette Scores and Elbow curve

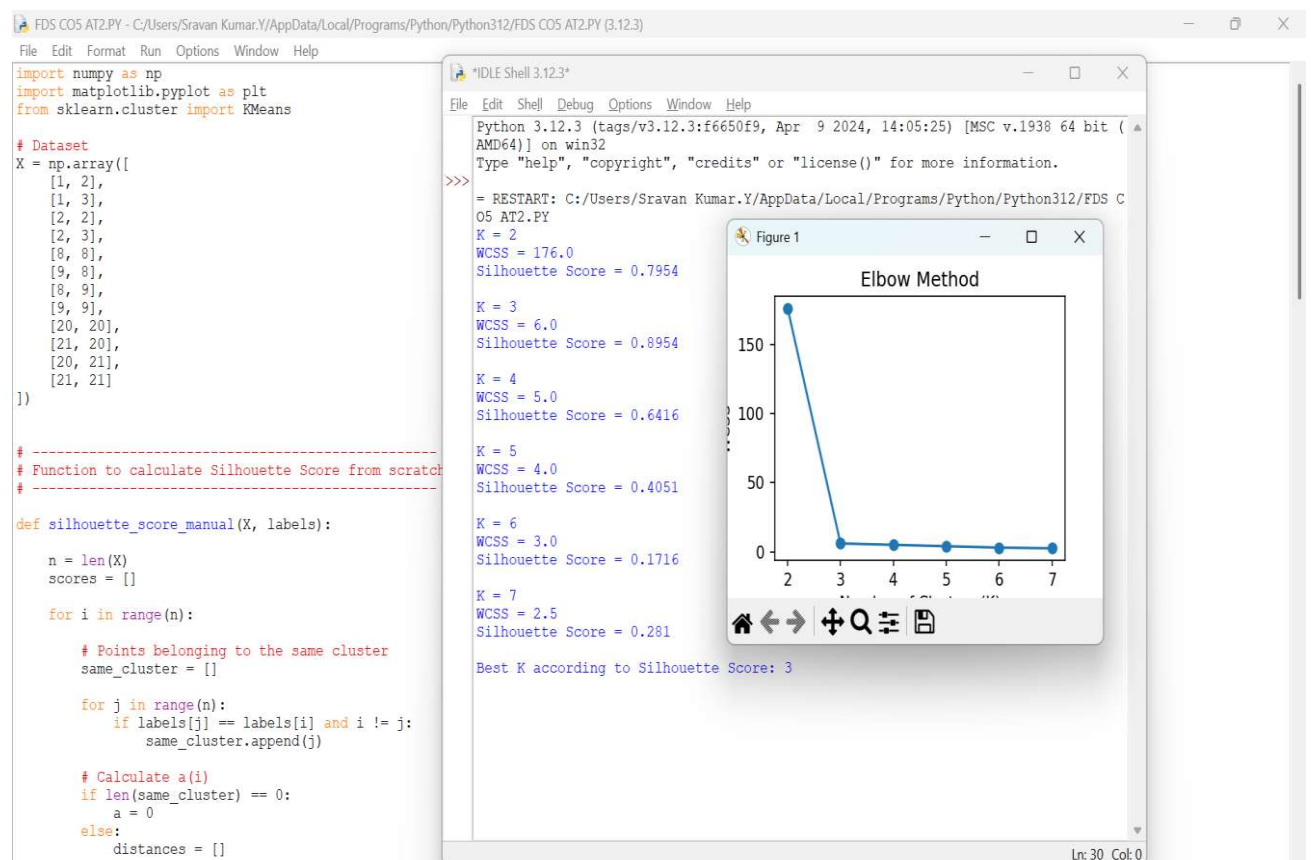
Select the K with better cluster quality

and meaningful separation

Display optimal K

STOP

Code



It also produces **two** graphs:

1. **Elbow Method graph** → K vs WCSS
2. **Silhouette graph** → K vs Silhouette Score

How the Silhouette Score works

For every data point:

a(i) = average distance from the point to other points in the **same cluster**.

b(i) = smallest average distance from the point to points in the **nearest different cluster**.

Then:

$$\text{Silhouette Score} = \frac{(b - a) / \max(a, b)}$$

The score is between **-1 and +1**.

- **Close to +1** → good clustering
- **Around 0** → overlapping clusters
- **Negative** → point may be assigned to the wrong cluster

The final score is the **average silhouette value of all data points**.

Reconciling Elbow and Silhouette

Suppose the results are:

$$\begin{array}{llll} \text{Elbow Method} & \rightarrow & K & = & 3 \\ \text{Silhouette Score} & \rightarrow & K & = & 3 \end{array}$$

Then:

Select K = 3 because both methods agree.

If they disagree:

$$\begin{array}{llll} \text{Elbow Method} & \rightarrow & K & = & 3 \\ \text{Silhouette Score} & \rightarrow & K & = & 4 \end{array}$$

Then compare the **silhouette scores** and the visual elbow. If $K = 4$ gives a substantially better silhouette score and produces meaningful clusters, choose **$K = 4$** . If the improvement is small and $K = 3$ gives a clearer elbow and simpler interpretation, choose **$K = 3$** .

Result

Thus, the optimal number of clusters was determined using both the Elbow Method and a manually calculated Silhouette Score, and the results were reconciled to select the most suitable value of k .